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D20A- 32

Experiment: 08
Configuring a Cryptocurrency Wallet Using MetaMask and Integrating It with a DApp

1. Aim
To configure a cryptocurrency wallet using MetaMask, connect it to a Decentralized Application (DApp), and understand the process of managing blockchain accounts, authorizing transactions, and interacting with smart contracts.

2. Theory
MetaMask is a cryptocurrency wallet that enables users to create and manage blockchain accounts, access digital assets, and interact with decentralized applications. It is available as a browser extension and a mobile application. MetaMask primarily supports Ethereum and other compatible blockchain networks.

A cryptocurrency wallet is a digital tool used to manage cryptographic keys associated with blockchain accounts. It does not directly store cryptocurrency like a physical wallet stores cash. Instead, it allows users to access and manage digital assets recorded on a blockchain.
A Decentralized Application (DApp) is an application that uses blockchain technology and smart contracts to provide services without relying entirely on a centralized authority. DApps are commonly used in decentralized finance, digital identity, blockchain gaming, NFT marketplaces, and digital asset management.

MetaMask serves as a bridge between the user and a blockchain-based application. When a user connects MetaMask to a DApp, the application can request access to the user's public wallet address. If a transaction is required, MetaMask displays the transaction details and asks the user to approve or reject it.

The major components involved in this experiment are:
- MetaMask Wallet: Provides an interface for managing blockchain accounts and approving transactions.
- Public Wallet Address: A unique address used to identify an account and receive digital assets.
- Private Key and Secret Recovery Phrase: Used to control access to the wallet. These must be kept confidential.
- Decentralized Application: Provides the interface through which users interact with blockchain functionality.

- Smart Contract: A program deployed on a blockchain that automatically executes predefined rules.
- Blockchain Network: A distributed network that validates transactions and records them on the blockchain.
- Gas Fees: Transaction fees paid for computational operations on networks that require gas, such as Ethereum.

When a user initiates a blockchain transaction through a DApp, the request is sent to MetaMask for authorization. After the user approves it, MetaMask signs the transaction using the account's cryptographic key. The signed transaction is submitted to the blockchain network for processing. Once confirmed, the resulting state change is recorded on the blockchain.

3. Requirements

1. A computer or laptop with a stable internet connection.
2. A supported web browser, such as Google Chrome.
3. MetaMask browser extension.
4. Access to a compatible decentralized application.
5. A supported blockchain network, preferably a test network such as Sepolia.
6. Test cryptocurrency, if required for transaction execution on the selected network.

4. Procedure

Step 1: Installing MetaMask

Open the official MetaMask website at https://metamask.io/. Download the browser extension for a supported browser and complete the installation. Open the extension from the browser toolbar.

Step 2: Creating a New Wallet

Select the option to create a new wallet. Follow the setup instructions, configure a strong password where applicable, and carefully back up the Secret Recovery Phrase. Complete the verification process to activate the wallet.

Step 3: Understanding the Wallet Interface

After setup, examine the MetaMask interface. Identify the account name, public wallet address, selected network, and available balance. The public address is used to identify the account, while the private key and recovery phrase provide control over it.

Step 4: Selecting a Blockchain Network

Open the network selector in MetaMask and choose a compatible blockchain network. For academic experimentation, use a supported test network such as

Sepolia. If the test network is not visible, enable test networks in the wallet settings, where supported.

Step 5: Accessing a Decentralized Application
Open a compatible DApp in a new browser tab. Locate the Connect Wallet button, which allows the application to establish a connection with a supported cryptocurrency wallet.

Step 6: Connecting MetaMask to the DApp
Click the wallet connection option and select MetaMask. A connection request will appear in the MetaMask extension. Review the request and approve it only after verifying that the website is legitimate.

Step 7: Verifying the Connection
After approval, return to the DApp and verify that the connected wallet address is displayed correctly. The application can now request the public account information permitted by the wallet connection.

Step 8: Interacting with the DApp
If the application provides a supported feature, such as reading smart contract data, submitting a test transaction, or interacting with a smart contract, perform the operation using the available interface. Review the details before approving any transaction.

Step 9: Authorizing a Transaction
When a transaction is initiated, MetaMask displays the transaction request, which may include the recipient, network, transaction value, and estimated gas fee. Examine the information and approve or reject the request as appropriate. Use only test assets for this experiment.

Step 10: Verifying the Transaction
If a transaction is submitted, observe its status in MetaMask. Where applicable, open the transaction hash in the relevant blockchain explorer to examine details such as the sender, recipient, block number, and confirmation status.

Step 11: Disconnecting the Wallet
After completing the experiment, disconnect the wallet from the DApp using its interface or MetaMask's connected-sites settings. This helps manage which applications have access to the wallet account.

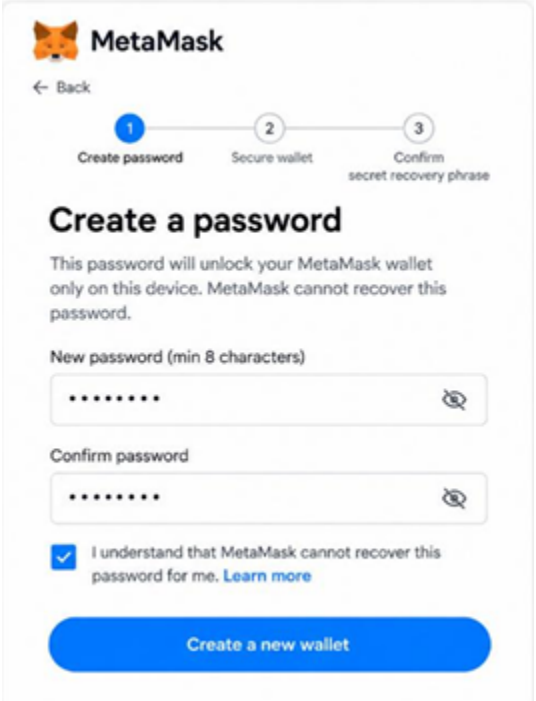
5. Observations

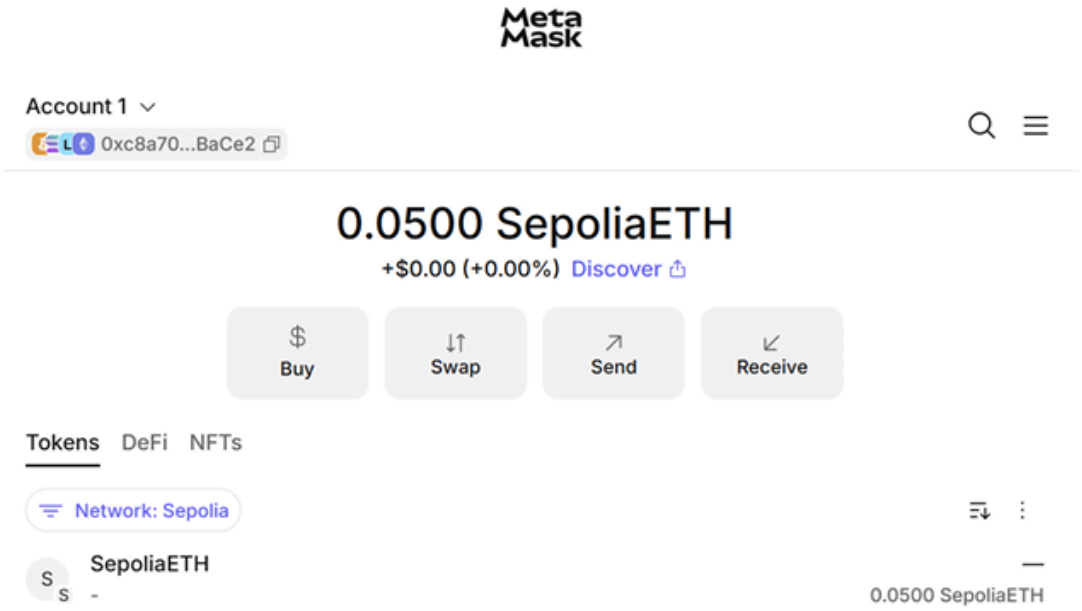
Parameter	Observation

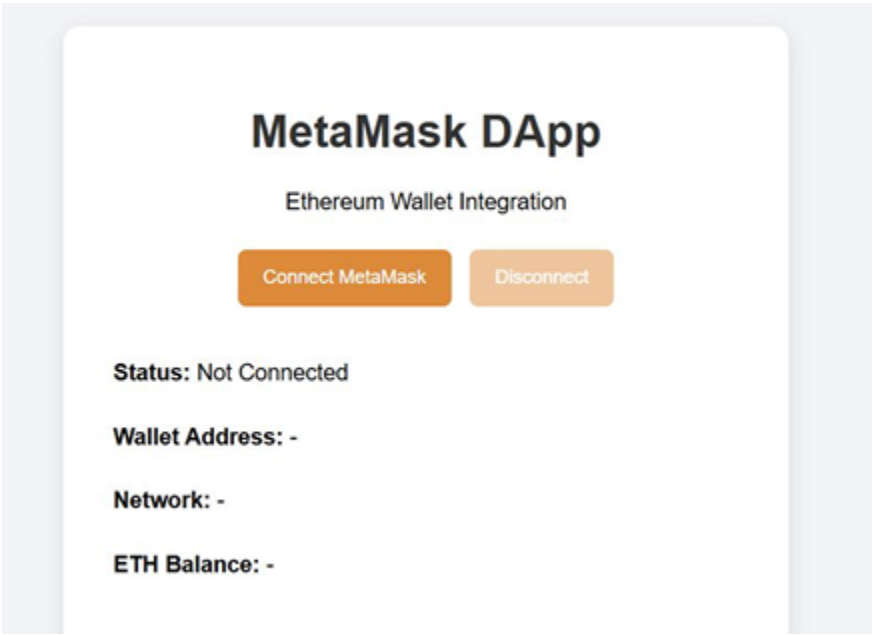
Wallet application	MetaMask
Wallet account	Created successfully
Public address	Generated for the account
Blockchain network	Selected a compatible network
DApp connection	Established after user authorization
Account display	Public wallet address visible in the DApp
Transaction authorization	Reviewed and approved or rejected through MetaMask
Transaction verification	Checked through wallet status or a blockchain explorer, if a transaction was submitted

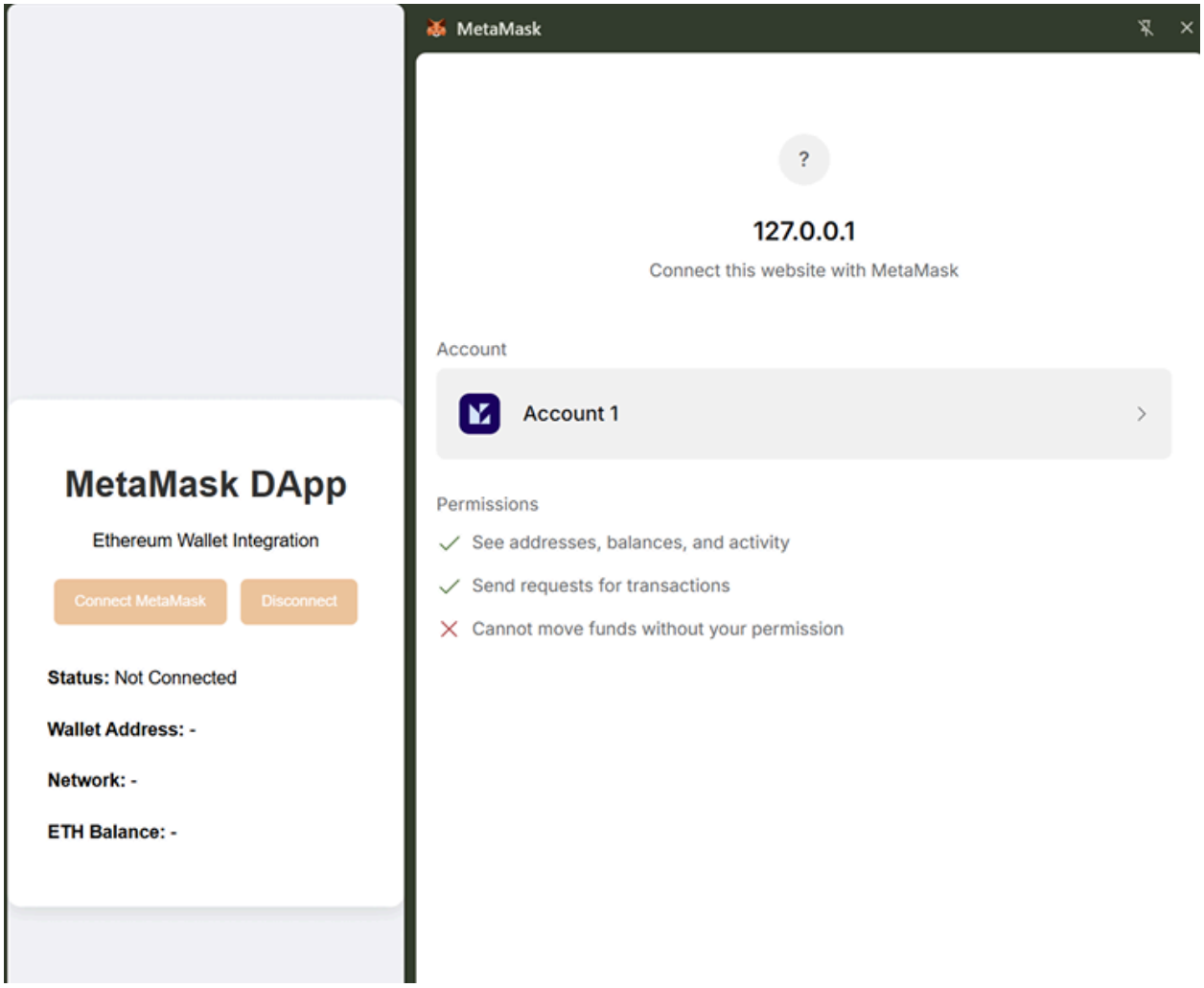
6. Output

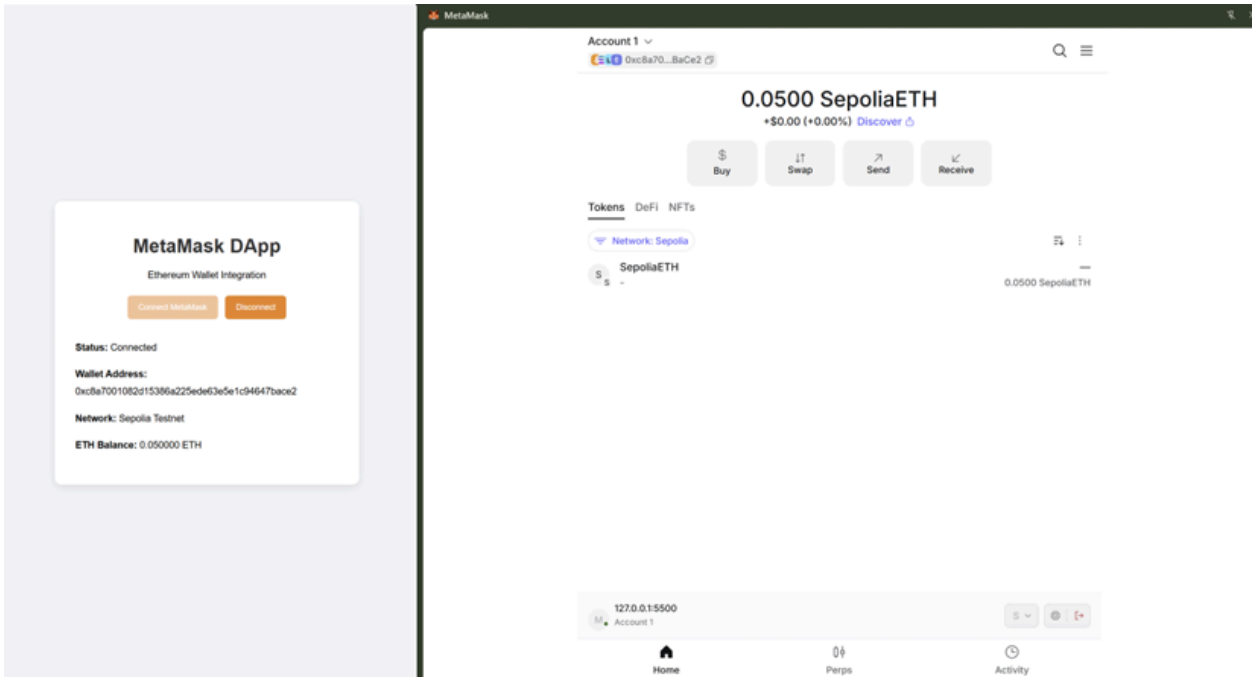
The MetaMask wallet was configured, and a blockchain account was created successfully. The wallet was connected to a compatible decentralized application, allowing the DApp to access the authorized public account information. The wallet interface demonstrated how transaction requests are reviewed and authorized. Where a test transaction was performed, its status could be verified through MetaMask or a blockchain explorer.



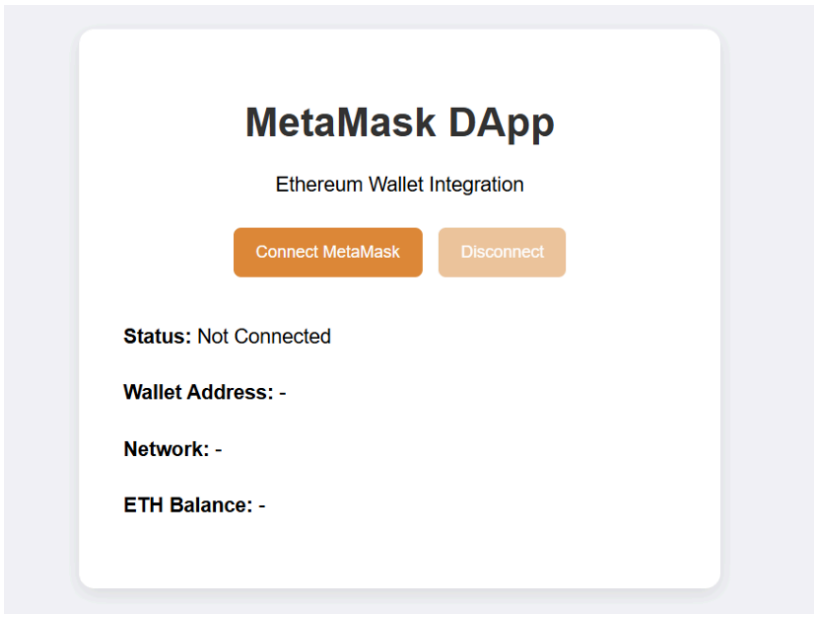

Meta Mask
Account 1
0xc8a70...BaCe2
0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
-
—
0.0500 SepoliaETH


MetaMask DApp
Ethereum Wallet Integration
Connect MetaMask
Disconnect
Status: Not Connected
Wallet Address: -
Network: -
ETH Balance: -


MetaMask DApp
Ethereum Wallet Integration
Connect MetaMask
Disconnect
Status: Not Connected
Wallet Address: -
Network: -
ETH Balance: -
MetaMask
127.0.0.1
Connect this website with MetaMask
Account
Account 1
Permissions
See addresses, balances, and activity
Send requests for transactions
Cannot move funds without your permission


MetaMask DApp
Ethereum Wallet Integration
Connect MetaMask
Disconnect
Status: Connected
Wallet Address:
0xc8a7001082d15386a225ede63e5e1c94647bace2
Network: Sepolia Testnet
ETH Balance: 0.050000 ETH
MetaMask
Account 1
0xc8a70...BaCe2
0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
0.0500 SepoliaETH
127.0.0.1:5500
Account 1
Home
Perps
Activity





7. Result

The experiment demonstrated the configuration of a cryptocurrency wallet using MetaMask and its integration with a decentralized application. The process provided practical knowledge of wallet creation, public addresses, blockchain network selection, wallet connectivity, transaction signing, and smart contract interaction.

It also demonstrated the role of user authorization in blockchain transactions and the importance of securing private keys and recovery phrases when using decentralized applications.

8. Precautions

1. Install MetaMask only from its official website or a verified browser extension listing.
2. Keep the Secret Recovery Phrase and private keys confidential and store them securely.
3. Verify the website address before connecting the wallet to a DApp.
4. Review transaction details, including the recipient, network, and gas fees, before approval.
5. Use a test network and test cryptocurrency for academic experimentation.

6. Never enter the Secret Recovery Phrase into a DApp or an unfamiliar website.
7. Disconnect from applications that are no longer in use and periodically review connected sites.
8. Remember that disconnecting a wallet does not automatically revoke previously granted smart contract token allowances; those permissions may need to be reviewed separately.